

Answer Key & Teacher Notes

Session 1 – Area of Regular Polygons · Session 2 – Composite Figures

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

TALK PROMPTS

- How do you know every center triangle has exactly the same area?
- Why is finding one triangle enough to find the area of the whole sign?
- Where does each triangle's height (the apothem) start and end?

WATCH FOR

- Finding the area of only one center triangle and stopping.
- Forgetting the $\frac{1}{2}$ in the triangle area (side $\times$ apothem $\times$ n).
- Using the distance from the center to a vertex instead of the apothem as the height.

PARTNER TASK

Decagon Tabletop. A regular decagon has sides of 6 cm and an apothem of about 9 cm. One partner finds one center triangle (27 cm^2); the other multiplies by the number of triangles (270 cm^2). Switch roles for the octagon stepping stone.

SESSION 2 • TEACHER NOTES

TALK PROMPTS

- When do you add the pieces and when do you subtract? How do you decide?
- How can you split the L-shaped floor a different way and still get the same area?

WATCH FOR

- Adding the area of a cut-out (window, vent, notch) instead of subtracting it.
- Forgetting the $\frac{1}{2}$ when a piece is a triangle.
- Using the wrong side length for a piece of an L shape (counting an overlap twice).

PARTNER TASK

Tool Shed Side. A shed side is a rectangle 12 ft by 7 ft with a triangle top (base 12 ft, height 4 ft). One partner finds the rectangle (84 ft^2), the other the triangle (24 ft^2); together decide to add (108 ft^2). Then do the banner: 18 in by 12 in with a triangle (base 6 in, height 12 in) cut away: $216 - 36 = 180 \text{ in}^2$.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.1	1	1	1	1	1	2	3	2	3	1	1
S1 • Form B	6.G.A.1	1	1	1	1	1	2	3	2	3	1	1
S2 • Form A	6.G.A.1	1	1	1	1	1	2	3	3	3	1	1
S2 • Form B	6.G.A.1	1	1	1	1	1	2	3	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

regular polygon: All sides **equal** and all angles equal.

congruent: Exactly the same **size** and shape.

apothem: Distance from the **center** to the middle of a side.

- Lines from the center to each vertex make **congruent** triangles.
- A regular polygon with n sides makes n center triangles.
- Area = $n \times (\frac{1}{2} \times \text{side} \times \text{apothem})$.

Example: 5 sides make **5** center triangles. One triangle: $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$. All triangles: $5 \times 35 = 175 \text{ cm}^2$.

Answer: Area $\approx 175 \text{ cm}^2$

YOU TRY

1. One triangle $\frac{1}{2} \times 4 \times 3.5 = 7 \text{ in}^2$; $6 \times 7 \approx 42 \text{ in}^2$
2. a. 8 b. 10

EXIT TICKET

1. a. 5 b. $\frac{1}{2} \times 8 \times 5 = 20 \text{ ft}^2$
2. One triangle $\frac{1}{2} \times 10 \times 8.7 = 43.5 \text{ cm}^2$; $6 \times 43.5 \approx 261 \text{ cm}^2$
3. **C** (120 in^2)

PRACTICE FORM A

1. a. $\frac{1}{2} \times 10 \times 7 = 35 \text{ cm}^2$ b. $\frac{1}{2} \times 16 \times 11 = 88 \text{ in}^2$ c. $\frac{1}{2} \times 7 \times 8 = 28 \text{ m}^2$ d. $\frac{1}{2} \times 6 \times 9 = 27 \text{ ft}^2$
2. a. 6 b. $\frac{1}{2} \times 9 \times 7.8 = 35.1 \text{ ft}^2$ c. $6 \times 35.1 \approx 210.6 \text{ ft}^2$
3. One triangle $\frac{1}{2} \times 16 \times 11 = 88 \text{ in}^2$; $5 \times 88 \approx 440 \text{ in}^2$
4. One triangle $\frac{1}{2} \times 7 \times 8 = 28 \text{ cm}^2$; $8 \times 28 \approx 224 \text{ cm}^2$
5. One triangle $\frac{1}{2} \times 24 \times 21 = 252 \text{ ft}^2$; $6 \times 252 \approx 1,512 \text{ ft}^2$
6. One triangle $\frac{1}{2} \times 18 \times 12.4 = 111.6 \text{ m}^2$; $5 \times 111.6 \approx 558 \text{ m}^2$
7. No. 45 cm^2 is only one center triangle. The tile has 6 triangles: $6 \times 45 = 270 \text{ cm}^2$.
8. a. $6 \times (\frac{1}{2} \times 8 \times 6.9) = 6 \times 27.6 = 165.6 \text{ cm}^2$ b. $25 \times 165.6 = 4,140 \text{ cm}^2$
9. Lena forgot the $\frac{1}{2}$ in each triangle's area. One triangle is $\frac{1}{2} \times 22 \times 15 = 165 \text{ in}^2$, so $A = 5 \times 165 = 825 \text{ in}^2$.
10. **C** (6)
11. **C** (100 cm^2)

Self-check word POLYGON

PRACTICE FORM B

1. a. $\frac{1}{2} \times 12 \times 8 = 48 \text{ cm}^2$ b. $\frac{1}{2} \times 14 \times 9 = 63 \text{ in}^2$ c. $\frac{1}{2} \times 5 \times 6 = 15 \text{ m}^2$ d. $\frac{1}{2} \times 9 \times 10 = 45 \text{ ft}^2$
2. a. 6 b. $\frac{1}{2} \times 12 \times 10.4 = 62.4 \text{ ft}^2$ c. $6 \times 62.4 \approx 374.4 \text{ ft}^2$
3. One triangle $\frac{1}{2} \times 14 \times 9.6 = 67.2 \text{ in}^2$; $5 \times 67.2 \approx 336 \text{ in}^2$
4. One triangle $\frac{1}{2} \times 6 \times 7.2 = 21.6 \text{ cm}^2$; $8 \times 21.6 \approx 172.8 \text{ cm}^2$
5. One triangle $\frac{1}{2} \times 20 \times 17.3 = 173 \text{ ft}^2$; $6 \times 173 \approx 1,038 \text{ ft}^2$
6. One triangle $\frac{1}{2} \times 12 \times 8.3 = 49.8 \text{ m}^2$; $5 \times 49.8 \approx 249 \text{ m}^2$
7. No. 28 cm^2 is only one center triangle. The tile has 6 triangles: $6 \times 28 = 168 \text{ cm}^2$.
8. a. $6 \times (\frac{1}{2} \times 6 \times 5.2) = 6 \times 15.6 = 93.6 \text{ cm}^2$ b. $30 \times 93.6 = 2,808 \text{ cm}^2$
9. Mateo forgot the $\frac{1}{2}$ in each triangle's area. One triangle is $\frac{1}{2} \times 10 \times 12 = 60 \text{ in}^2$, so $A = 8 \times 60 = 480 \text{ in}^2$.
10. **C** (8)
11. **B** (number of sides)

Self-check word HEXAGON

GUIDED NOTES

composite figure: A shape made of two or more **simpler** shapes.

decompose: To **break** a shape into smaller shapes you know.

subtract: To **take away** the area of a piece that is cut out.

- Split a composite figure into rectangles, triangles, or other shapes you **know**.
- Add** the areas of pieces that are joined together.
- Subtract** the area of a piece that is cut out.

Example: Whole rectangle: $14 \times 9 = 126 \text{ cm}^2$. Cut-out triangle: $\frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$. The triangle is removed, so **subtract**. Answer: $126 - 12 = 114 \text{ cm}^2$

YOU TRY

- Bottom $8 \times 3 = 24 \text{ m}^2$; top $3 \times 4 = 12 \text{ m}^2$; total 36 m^2
- a. subtract b. add

EXIT TICKET

- a. $50 - 6 = 44 \text{ ft}^2$ b. $24 + 8 = 32 \text{ m}^2$
- Rectangle $10 \times 6 = 60 \text{ ft}^2$; triangle $\frac{1}{2} \times 10 \times 4 = 20 \text{ ft}^2$; total 80 ft^2
- C** (132 ft^2)

PRACTICE FORM A

- a. add: 52 cm^2 b. subtract: 84 in^2 c. subtract: $81 - 9 = 72 \text{ ft}^2$ d. subtract: $40 - 4 = 36 \text{ m}^2$
- a. $20 \times 14 = 280 \text{ in}^2$ b. $\frac{1}{2} \times 14 \times 5 = 35 \text{ in}^2$ c. $280 - 35 = 245 \text{ in}^2$
- $13 \times 10 = 130$; $5 \times 11 = 55$; total 185 ft^2
- $22 \times 10 = 220$; $220 - 25 = 195 \text{ ft}^2$
- $24 \times 9 = 216$; $216 - 18 = 198 \text{ ft}^2$
- Rectangle $26 \times 15 = 390 \text{ ft}^2$; roof $\frac{1}{2} \times 26 \times 11 = 143 \text{ ft}^2$; total 533 ft^2
- No. Deja forgot to subtract the corner. $800 - 100 = 700 \text{ in}^2$ is left.
- a. $30 \times 20 = 600 \text{ m}^2$ b. $600 - \frac{1}{2} \times 12 \times 10 = 600 - 60 = 540 \text{ m}^2$ c. $540 \times 14 = \$7,560$
- Omar added the vent, but the vent is not painted, so its area must be subtracted. Painted area = $209 - 25 = 184 \text{ ft}^2$.
- B** (subtract its area)
- C** (two or more simpler shapes)

Self-check word **SUBTRACT**

PRACTICE FORM B

- a. add: 46 cm^2 b. subtract: 56 in^2 c. subtract: $64 - 4 = 60 \text{ ft}^2$ d. subtract: $60 - 9 = 51 \text{ m}^2$
- a. $24 \times 16 = 384 \text{ in}^2$ b. $\frac{1}{2} \times 16 \times 6 = 48 \text{ in}^2$ c. $384 - 48 = 336 \text{ in}^2$
- $12 \times 9 = 108$; $4 \times 10 = 40$; total 148 ft^2
- $18 \times 9 = 162$; $162 - 16 = 146 \text{ ft}^2$
- $20 \times 8 = 160$; $160 - 15 = 145 \text{ ft}^2$
- Rectangle $22 \times 14 = 308 \text{ ft}^2$; roof $\frac{1}{2} \times 22 \times 10 = 110 \text{ ft}^2$; total 418 ft^2
- No. Nico forgot to subtract the corner. $720 - 64 = 656 \text{ in}^2$ is left.
- a. $28 \times 18 = 504 \text{ m}^2$ b. $504 - \frac{1}{2} \times 10 \times 8 = 504 - 40 = 464 \text{ m}^2$ c. $464 \times 12 = \$5,568$
- Lila added the vent, but the vent is not painted, so its area must be subtracted. Painted area = $170 - 16 = 154 \text{ ft}^2$.
- A** (55 in^2)
- C** (65 cm^2)

Self-check word **COMBINED**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · AREA OF REGULAR POLYGONS

1. One triangle $\frac{1}{2} \times 10 \times 12.1 = 60.5$; $8 \times 60.5 \approx 484 \text{ m}^2$
2. Hexagon: $P = 24 \text{ cm}$, $A \approx \frac{1}{2} \times 24 \times 3.5 = 42 \text{ cm}^2$;
Pentagon: $P = 50 \text{ ft}$, $A \approx \frac{1}{2} \times 50 \times 6.9 = 172.5 \text{ ft}^2$
3. a. $\approx \frac{1}{2} \times 30 \times 4.1 = 61.5 \text{ cm}^2$ b. $\approx \frac{1}{2} \times 60 \times 8.7 = 261 \text{ ft}^2$
4. Area $\approx 6 \times (\frac{1}{2} \times 6 \times 5.2) = 93.6 \text{ ft}^2$; cost $\approx 93.6 \times \$3 = \280.80
5. He found only one of the 5 triangles. Area $\approx 5 \times 22 = 110 \text{ cm}^2$.

SHEET 2 · COMPOSITE FIGURES

1. Bottom $10 \times 4 = 40$; top $4 \times 4 = 16$; total 56 m^2
2. a. subtract (wall area – window area) b. add (room area + closet area)
3. Top bar $10 \times 2 = 20$; stem $4 \times 4 = 16$; total 36 in^2
4. Rectangle $12 \times 8 = 96$; notch base $12 - 3 - 3 = 6$, $\frac{1}{2} \times 6 \times 4 = 12$; $96 - 12 = 84 \text{ cm}^2$
5. $240 - 25 = 215 \text{ ft}^2$
6. **A** (50 m^2)
7. **C** (46 in^2)